

## Homework 7

1. Suppose  $0 < a < b$ . Find the area of the torus obtained by revolving the circle  $(x-b)^2 + z^2 = a^2$  in the  $xz$ -plane about the  $z$  axis. (Hint: The torus may be parametrized by  $x = (b + a \cos \phi) \cos \theta$ ,  $y = (b + a \cos \phi) \sin \theta$ ,  $z = a \sin \phi$ , with  $0 \leq \phi, \theta \leq 2\pi$ . )
2. Use Stokes's theorem to evaluate  $\int_C [ydx + y^2dy + (x + 2z)dz]$  where  $C$  is the curve of intersection of sphere  $x^2 + y^2 + z^2 = a^2$  and the plane  $y + z = a$ , oriented counterclockwise as viewed from above.
3. Suppose  $a_n > 0$  and  $\sum a_n$  converges. Put  $r_n = \sum_{m=n}^{\infty} a_m$ . Prove that  $\frac{a_m}{r_m} + \cdots + \frac{a_n}{r_n} > 1 - \frac{r_n}{r_m}$  if  $m < n$ , and deduce that  $\sum \frac{a_n}{r_n}$  diverges.
4. Suppose  $\{p_n\}$  and  $\{q_n\}$  are Cauchy sequences in a metric space  $\mathbf{X}$ . Show that the sequence  $\{d(p_n, q_n)\}$  converges.
5. If the sequence  $\{x_n\}_{n \geq 0}$  converges to  $x$ , define a new sequence  $\{y_n\}_{n \geq 0}$  by

$$y_n = \frac{x_0 + x_1 + \cdots + x_n}{n + 1}$$

Show that  $\lim_{n \rightarrow \infty} y_n = x$ .